

Guide to Problem 1a: Compensated Demand for Perfect Substitutes

ECO 310 Problem Set 2

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1 Problem Statement

Calculate the compensated demand bundle $(x_1^c(p_1, p_2, u^*), x_2^c(p_1, p_2, u^*))$ for perfect substitute preferences with $u(x_1, x_2) = x_1 + x_2$.

Hint: You should have a piecewise function, depending on which good is cheaper.

2 Key Concepts

2.1 Perfect Substitute Preferences

With utility function $u(x_1, x_2) = x_1 + x_2$:

- The goods are **perfect substitutes** - the consumer is willing to trade them one-for-one
- Indifference curves are **straight lines** with slope -1
- The MRS (marginal rate of substitution) is constant: $MRS = -1$
- Solutions are typically **corner solutions** - buy only one good

Intuition

If you like apples and oranges equally, and apples are cheaper, why would you ever buy oranges? You'd spend all your money on apples!

2.2 Compensated (Hicksian) Demand

Compensated demand solves the **expenditure minimization problem**:

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \tag{1}$$

subject to:

$$x_1 + x_2 \geq u^* \quad (\text{utility constraint}) \tag{2}$$

$$x_1, x_2 \geq 0 \quad (\text{non-negativity}) \tag{3}$$

This asks: **What's the cheapest way to reach utility level u^* ?**

Key Difference

Regular (Marshallian) demand: Maximize utility subject to budget constraint

Compensated (Hicksian) demand: Minimize expenditure subject to utility constraint

3 Solving the Problem

3.1 Step 1: Set Up the Problem

We want to minimize $p_1x_1 + p_2x_2$ subject to $x_1 + x_2 \geq u^*$.

Since we want to reach exactly u^* at minimum cost, the utility constraint will bind:

$$x_1 + x_2 = u^* \quad (4)$$

3.2 Step 2: Use the Constraint

From the utility constraint: $x_2 = u^* - x_1$

Substitute into the objective function:

$$\text{Cost} = p_1x_1 + p_2(u^* - x_1) = p_1x_1 + p_2u^* - p_2x_1 = (p_1 - p_2)x_1 + p_2u^* \quad (5)$$

3.3 Step 3: Minimize Cost

The cost function is **linear** in x_1 :

$$\text{Cost} = (p_1 - p_2)x_1 + p_2u^* \quad (6)$$

- If $p_1 - p_2 < 0$ (i.e., $p_1 < p_2$): coefficient of x_1 is negative, so increase x_1 as much as possible
- If $p_1 - p_2 > 0$ (i.e., $p_1 > p_2$): coefficient of x_1 is positive, so decrease x_1 as much as possible
- If $p_1 = p_2$: cost doesn't depend on x_1 , so any combination works

3.4 Step 4: Apply Non-Negativity Constraints

Remember: $x_1 \geq 0$ and $x_2 \geq 0$

From $x_2 = u^* - x_1 \geq 0$, we get $x_1 \leq u^*$

So: $0 \leq x_1 \leq u^*$

3.5 Step 5: Find Corner Solutions

Case 1: If $p_1 < p_2$ (good 1 is cheaper)

- Maximize x_1 : set $x_1 = u^*$
- Then $x_2 = u^* - x_1 = 0$
- Buy only good 1 (the cheaper good)

Case 2: If $p_1 > p_2$ (good 2 is cheaper)

- Minimize x_1 : set $x_1 = 0$

- Then $x_2 = u^* - 0 = u^*$
- Buy only good 2 (the cheaper good)

Case 3: If $p_1 = p_2$ (goods cost the same)

- Any combination where $x_1 + x_2 = u^*$ minimizes cost
- Could choose $x_1 \in [0, u^*]$ and $x_2 = u^* - x_1$

4 Final Answer

The compensated demand bundle is:

Compensated Demand for Perfect Substitutes

$$(x_1^c(p_1, p_2, u^*), x_2^c(p_1, p_2, u^*)) = \begin{cases} (u^*, 0) & \text{if } p_1 < p_2 \\ (0, u^*) & \text{if } p_1 > p_2 \\ \text{any } (x_1, u^* - x_1) \text{ with } x_1 \in [0, u^*] & \text{if } p_1 = p_2 \end{cases} \quad (7)$$

5 Economic Intuition

1. **Perfect substitutes $\Rightarrow$ corner solutions:** Since goods are equally valuable per unit consumed, always buy the cheaper one
2. **Why it's "piecewise":** The optimal choice discontinuously jumps from all good 1 to all good 2 as relative prices change
3. **The u^* appears:** We need to consume exactly u^* total units to reach the target utility level
4. **Comparison to Marshallian demand:** Regular demand depends on income m ; compensated demand depends on target utility u^*

6 Connection to Part (b): Expenditure Function

Once you have compensated demand, the expenditure function is straightforward:

$$e(p_1, p_2, u^*) = p_1 \cdot x_1^c + p_2 \cdot x_2^c \quad (8)$$

Substituting our answer:

- If $p_1 < p_2$: $e = p_1 \cdot u^* + p_2 \cdot 0 = p_1 u^*$
- If $p_1 > p_2$: $e = p_1 \cdot 0 + p_2 \cdot u^* = p_2 u^*$
- If $p_1 = p_2$: $e = p_1 u^*$ (or equivalently $p_2 u^*$)

This simplifies to: $e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^*$

7 Graphical Representation

Imagine a graph with x_1 on the horizontal axis and x_2 on the vertical axis:

- **Budget line:** $p_1x_1 + p_2x_2 = E$ (slope = $-p_1/p_2$)
- **Indifference curve:** $x_1 + x_2 = u^*$ (slope = -1)

The minimum expenditure point is where:

- If budget line is steeper than IC ($-p_1/p_2 < -1$, i.e., $p_1 < p_2$): corner at $(u^*, 0)$
- If budget line is flatter than IC ($-p_1/p_2 > -1$, i.e., $p_1 > p_2$): corner at $(0, u^*)$
- If slopes are equal: entire segment of IC from $(0, u^*)$ to $(u^*, 0)$ is optimal